Vol 12 Chapter 3 — Molecular Orbital Theory

Keeper (author pass — deep math/physics revision)

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Chapter 3 — Molecular Orbital Theory

Level 1 — one sentence

Molecular orbital (MO) theory treats electrons in molecules as occupying delocalized orbitals spanning the whole molecule (LCAO-MO construction), with bonding/antibonding/non-bonding orbitals organizing molecular spectroscopy and reactivity — and modern computational quantum chemistry (DFT, post-HF) extends this to quantitative predictions for arbitrary molecules.

Level 2 — graduate-physicist precision

3.1 LCAO-MO

Molecular orbitals built from linear combinations of atomic orbitals:

$$\psi_{MO} = \sum_i c_i \phi_i^{AO}$$

For H_2: two atomic 1s orbitals ϕ_A, ϕ_B combine into: - Bonding $\sigma_g = (\phi_A + \phi_B)/\sqrt{2(1+S)}$: lower energy - Antibonding $\sigma_u^* = (\phi_A - \phi_B)/\sqrt{2(1-S)}$: higher energy

With overlap integral $S = \langle \phi_A | \phi_B \rangle$.

Two electrons fill σ_g, none in σ_u^*: bonding net.

3.2 Bond order

Bond order = (bonding − antibonding)/2.

H_2: bond order 1. He_2: bond order 0 (equal filling) → He_2 unstable.

For O_2: bond order 2 with 2 unpaired electrons in π^* — explains paramagnetism (one of MO's early triumphs over valence-bond theory).

3.3 Hückel theory for π systems

For planar conjugated systems (benzene, polyenes, porphyrins): π electrons in MOs built from p_z orbitals.

Benzene: 6 π MOs from 6 carbon $2p_z$. Lowest 3 (3 pairs of electrons): filled. Aromatic stabilization energy $\sim$ 36 kcal/mol.

Hückel 4n+2 rule: $4n + 2$ π electrons → aromatic (stable). Examples: benzene ($n = 1$), naphthalene ($n = 2$), [18]annulene ($n = 4$).

3.4 Density Functional Theory (DFT)

Kohn-Sham 1965 (Nobel 1998): the ground-state energy is a functional of the electron density alone. Exact in principle; approximate exchange-correlation functionals in practice (LDA, GGA, hybrid B3LYP, etc.).

DFT enables routine quantum chemistry on molecules with hundreds of atoms. Standard codes: Gaussian, VASP, Quantum ESPRESSO, NWChem.

3.5 Substrate-mechanism reading

MOs are substrate K-type configurations on multi-atom boundary conditions (Vol 9 Ch 3 band-structure analog at finite molecule).

Each MO is a substrate K-type with definite symmetry under molecular point group; filling Pauli-respecting.

3.6 K-audit anchors

- Vol 5 Ch 9 (Pauli, spin-statistics)
- Vol 9 Ch 3 (band structure)

Level 3 — 5th-grader accessibility

Molecular orbital theory: electrons in molecules live in orbitals spread across the whole molecule. LCAO-MO: combine atomic orbitals → molecular orbitals. Bonding (lower energy) vs antibonding (higher) vs non-bonding. H_2: 2 electrons fill bonding → bond. He_2: 4 electrons fill bonding + antibonding → no bond. Benzene aromatic (6 π electrons): special stability from Hückel 4n+2 rule. DFT: modern computational chemistry handles 100s of atoms via density-functional approach. In BST: MOs are substrate K-types on multi-atom configurations.

What comes next

Chapter 4 develops chemical reactions and thermodynamics.

Where to look this up

- Atkins-Friedman, *Molecular Quantum Mechanics*
- Parr-Yang, *Density-Functional Theory of Atoms and Molecules*
- BST: Vol 5 Ch 9; Vol 9 Ch 3